



Bluestock Mutual Fund Analytics Capstone

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Python

SQL

SQLite

Power BI

Pandas

NumPy

Executive Summary



Scalable ETL Pipeline

Built scalable ETL pipeline with SQLite storage



Risk-Adjusted Evaluation

Evaluated funds using Sharpe Ratio, Alpha, Beta, VaR



Interactive Dashboard

Developed 4-page interactive Power BI dashboard



Data-Driven Recommendations

Generated data-driven fund recommendations

⚠ Problem Statement



Fragmented Data

Mutual fund data is fragmented across multiple sources



No Unified Platform

No unified platform for risk-adjusted performance analysis



Manual & Error-Prone

Manual analysis is slow and error-prone



The Solution

Automated analytics pipeline + Power
BI dashboard

ETL

SQLite

Python

Power BI

Data Sources

10 Datasets Used

1

Fund Master
Dataset

2

NAV History
Dataset

3

AUM by Fund
House Dataset

4

Monthly SIP
Inflows Dataset

5

Category
Inflows Dataset

6

Industry Folio
Count Dataset

7

Scheme
Performance
Dataset

8

Investor
Transactions
Dataset

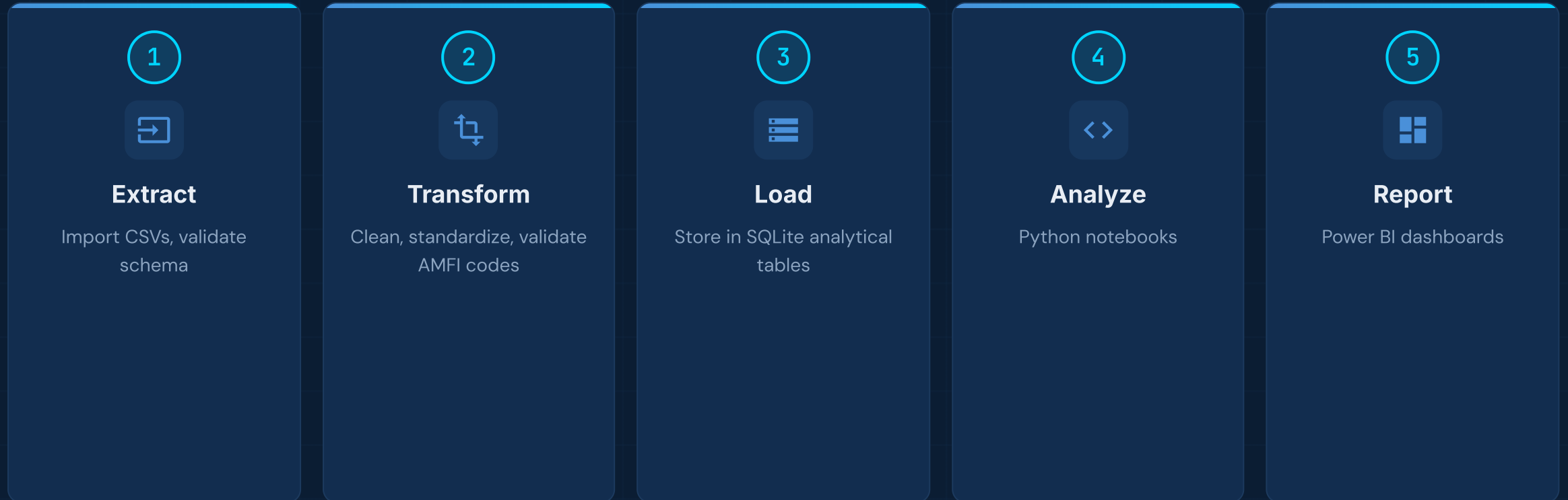
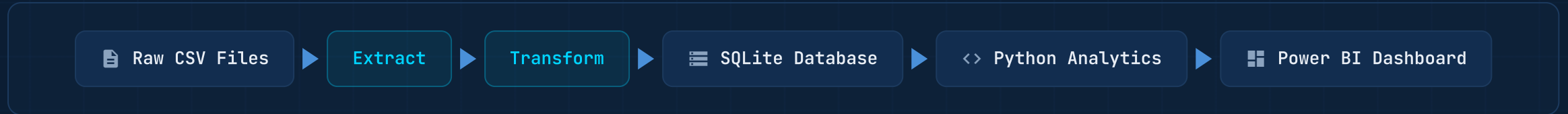
9

Portfolio
Holdings
Dataset

10

Benchmark
Indices Dataset

ETL Architecture



Data Cleaning

Tools: [Python](#) · [Pandas](#) · [NumPy](#)



7

Cleaning Steps

- ✓ 01 Removed duplicate records
- ✓ 02 Handled missing and null values
- ✓ 03 Standardized column names
- ✓ 04 Converted date formats
- ✓ 05 Corrected numeric data types
- ✓ 06 Validated unique AMFI codes
- ✓ 07 Cross-dataset consistency checks

Exploratory Data Analysis

Six key findings from the dataset



CATEGORY DOMINANCE

Equity Funds

Largest share of schemes



GROWTH TREND

SIP Inflows

Consistent growth trend



DISTRIBUTION

AUM Concentration

Concentrated among top fund houses



TOP FUND HOUSE

SBI Mutual Fund

Highest AUM (~9M crore)



INVESTOR PREFERENCE

Equity Folios

Highest in equity-oriented schemes



INDUSTRY GROWTH

Industry AUM Growth

Grew from 3M to 6M+ crore (2022–2025)

Performance Analytics

Risk-adjusted metrics and top performer identification



Sharpe Ratio

Identified top risk-adjusted performers



Sortino Ratio

Measured downside protection



Alpha

Positive values = superior active management



Beta

Moderate values = balanced market exposure



CAGR

Long-term compounding performance measured



TOP PERFORMER

SBI Small Cap Fund

23.39%

3yr return

Advanced Analytics

Risk quantification and concentration analysis



VaR & CVaR

Quantified maximum downside loss exposure across fund categories using Value at Risk and Conditional Value at Risk metrics.

 Downside Risk



Rolling 90-Day Sharpe

Fund efficiency varies by market cycle. Rolling analysis revealed dynamic risk-adjusted performance shifts over time.

 Dynamic Efficiency



HHI Index

Measured portfolio concentration risk using the Herfindahl-Hirschman Index to identify over-concentrated holdings.

 Concentration Metric



Diversification Insight

Lower HHI = better diversification. Some equity funds showed high concentration risk, flagging potential vulnerability.

 Portfolio Health



Investor Behavior Analysis

Demographics, transaction patterns, and risk indicators



Age 26–35

Highest investment amount among all age groups



Newer Cohorts

Investing larger amounts than older ones



SIP Continuity

Strong across most investors



At-Risk Flag

Average SIP gap greater than 35 days



Top States

Punjab and Tamil Nadu by investment amount



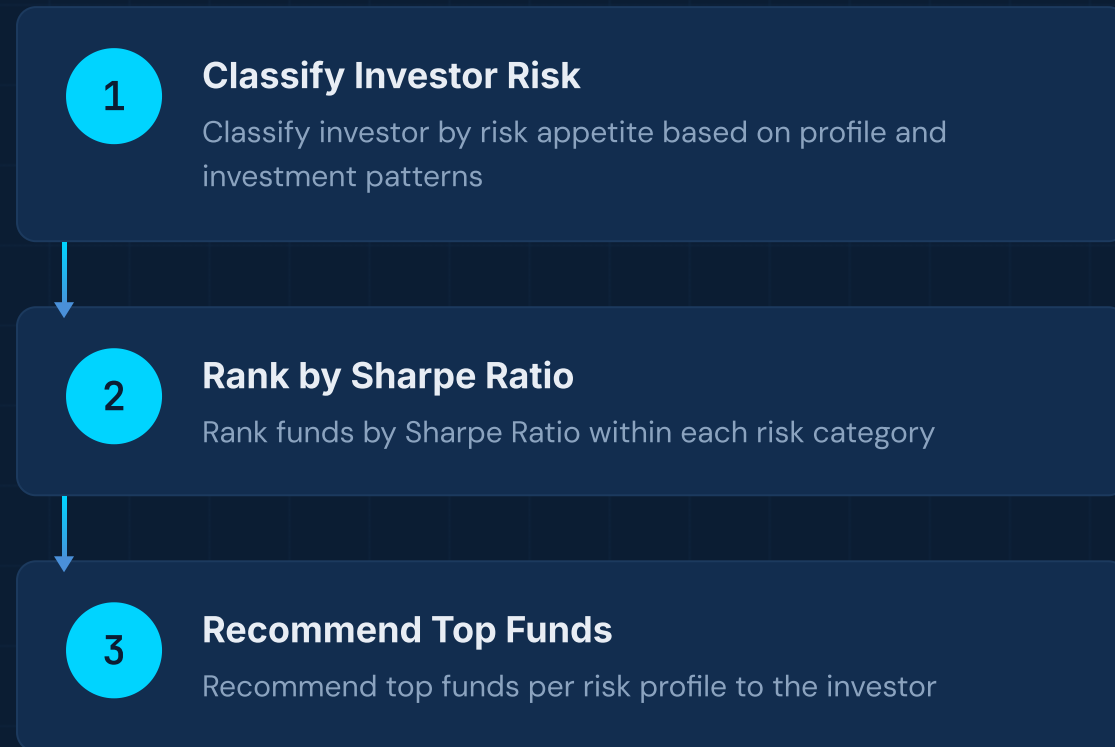
Transaction Split

58.49%	35.34%	6.17%
Lumpsum	Redemption	SIP

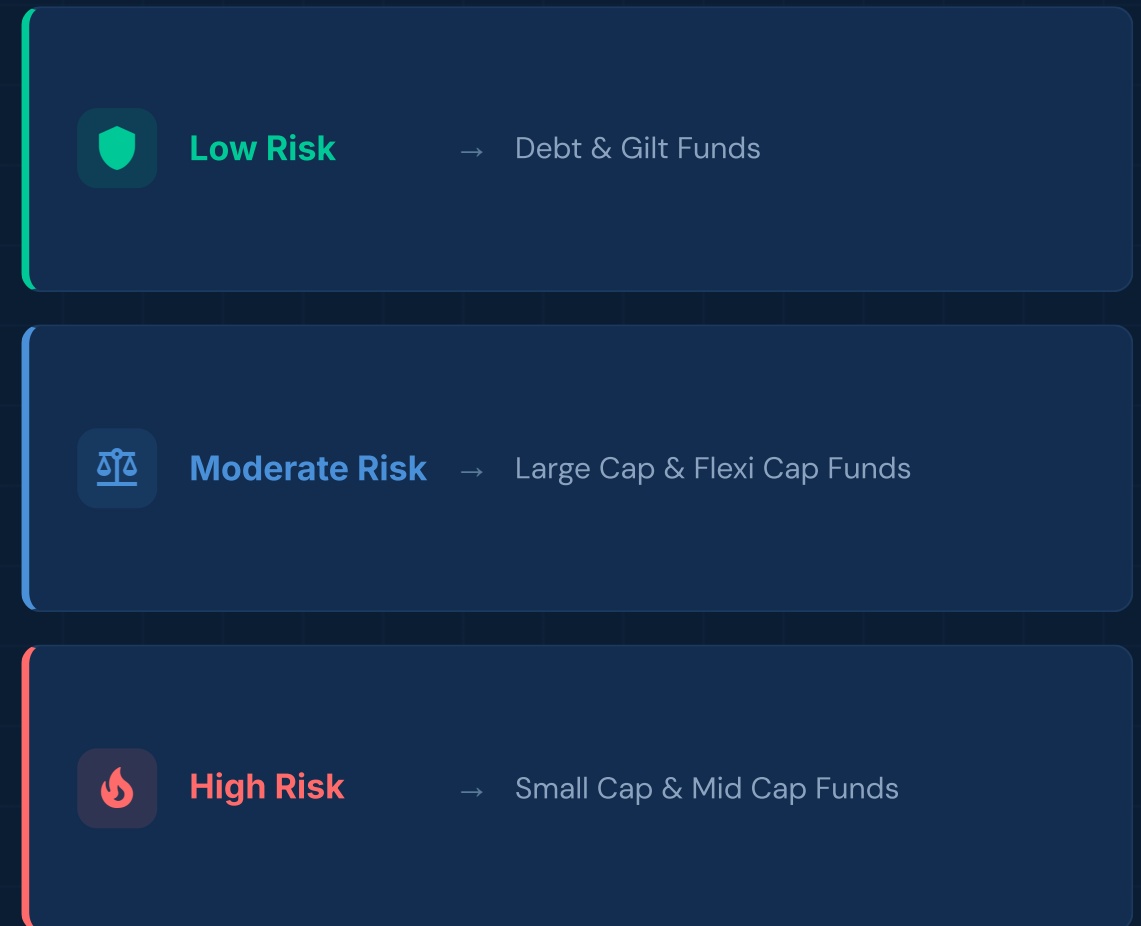


Fund Recommendation Engine

RECOMMENDATION PROCESS



RISK CATEGORIES



Power BI Dashboard

4 Dashboard Pages



Industry Overview

- AUM trends across the mutual fund industry
- Key performance indicators and benchmarks
- Fund house comparison and market share



Fund Performance

- NAV history and price movement tracking
- Returns analysis across time periods
- Risk grades and volatility assessment



Investor Analytics

- Demographics and age-based segmentation
- Transaction patterns and behavior analysis
- State-wise investment distribution



SIP & Market Trends

- SIP inflows and growth trajectory
- Category-wise market trend analysis
- Investment pattern shifts over time

Key Findings

Top 5 Insights

01

Hidden Downside Risk

High VaR/CVaR funds carry hidden downside risk that may not be apparent from returns alone

02

Non-Static Efficiency

Rolling Sharpe proves fund efficiency is not static — it varies significantly across market cycles

03

Newer Investors Invest More

Newer investor cohorts invest larger amounts compared to older investor groups

04

Strong SIP Continuity

SIP continuity is strong among retail investors, indicating disciplined investment behavior

05

Concentration Risk

Several equity funds have high concentration risk via HHI analysis, limiting diversification benefits

Limitations & Future Scope



Limitations



Static historical datasets only — no real-time or live market data integration



No macroeconomic variables included in the analysis framework



Rule-based recommendation engine without machine learning capabilities



Future Scope



Integrate real-time NAV APIs for live market data streaming



Deploy on cloud platform for scalable and accessible analytics



Build ML-based recommendation model for smarter fund suggestions



Add automated reporting and alerts for portfolio monitoring



Include sector-level indicators for deeper market analysis



This project built a complete **end-to-end** mutual fund analytics platform covering **data ingestion, cleaning, storage, performance analytics, and interactive dashboarding**.

<> github.com/rissam-insights/bluestock-mutual-fund-analytics-capstone

REFERENCES

AMFI India | NSE India

Bluestock Internship Guidelines

Python, Pandas, SQLite, Power BI Documentation

Thank You